

Competitive Programming

Section 62: Codeforces Course Ladder – Offline Problem Sheet

Competitive Programming Course

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1 Offline use

This is an original concise problem sheet, not a copy of Codeforces statements. It contains the information needed for an offline attempt and each official URL. Before submitting, read the official page: it controls any detail not present here. Ratings are approximate guideposts, not prerequisites.

Each rung also has an original, locally judgeable companion package under `problems/`; see [the companion index](#). The companion is a separate exercise for the same course skill, with skeletons, references, and tests. It is intentionally not a copy of the linked Codeforces statement.

2 Gate 1: foundations

2.1 1. 71A – Way Too Long Words (800)

Official: <https://codeforces.com/problemset/problem/71/A>

For each of at most 100 lowercase words (length at most 100), print it unchanged when its length is at most 10. Otherwise print its first character, the number of characters strictly between the ends, and its last character. Output one transformed word per input word.

2.2 2. 4C – Registration System (1300)

Official: <https://codeforces.com/problemset/problem/4/C>

Process up to 10^5 requested lowercase names. A first request for a name prints OK. A later request prints that base name followed by the smallest positive suffix that has not previously been issued, and the issued name also becomes occupied. Output one response per request.

2.3 3. 363B – Fence (1100)

Official: <https://codeforces.com/problemset/problem/363/B>

Given $n \leq 1.5 \cdot 10^5$ positive heights and k , choose k consecutive positions with minimum total height. Print a 1-indexed starting position; any minimum is accepted. The fence is a line, not a cycle.

2.4 4. 279B – Books (1400)

Official: <https://codeforces.com/problemset/problem/279/B>

Given at most 10^5 positive reading times and a time budget t , choose a contiguous block of books whose sum is at most t . Print the maximum possible number of books. An individual book may already exceed the budget.

2.5 5. 158B – Taxi (1100)

Official: <https://codeforces.com/problemset/problem/158/B>

Up to 10^5 groups, each of size 1 through 4, must ride in taxis of capacity 4. Groups cannot split, but different groups may share a taxi. Print the minimum taxi count.

2.6 6. 706B – Interesting Drink (1100)

Official: <https://codeforces.com/problemset/problem/706/B>

Store up to 10^5 drink prices. For each of up to 10^5 budget queries, print how many drinks cost no more than that budget. Prices and budgets fit in ordinary signed 32-bit input values; answers are counts.

3 Gate 2: graph, DP, math, and geometry

3.1 7. 520B – Two Buttons (1400)

Official: <https://codeforces.com/problemset/problem/520/B>

Starting from positive integer n , reach positive integer m using only $x \rightarrow 2x$ and $x \rightarrow x-1$. With $n, m \leq 10^4$, print the fewest operations. The intermediate values may exceed m , but a finite state bound must be justified.

3.2 8. 455A – Boredom (1500)

Official: <https://codeforces.com/problemset/problem/455/A>

From a multiset of at most 10^5 integers, repeatedly choose a value x , add x to the score, and remove every occurrence of $x-1$, x , and $x+1$. Maximize the total. Values are positive and bounded by 10^5 .

3.3 9. 300C – Beautiful Numbers (1800)

Official: <https://codeforces.com/problemset/problem/300/C>

Count length- n decimal strings whose digits are only a and b and whose digit sum, written in decimal, itself uses only a and b . The input has $1 \leq a < b \leq 9$ and $n \leq 10^6$, so factorials modulo $1_000_000_007$ are required. Print the count modulo that prime.

3.4 10. 166B – Polygons (2100, geometry challenge)

Official: <https://codeforces.com/problemset/problem/166/B>

You receive a strictly convex outer polygon A and an arbitrary simple inner polygon B , both listed clockwise with integer coordinates. Decide whether all of B lies strictly inside A . A vertex of B on A 's boundary is not allowed.

3.5 11. 20C – Dijkstra? (1900)

Official: <https://codeforces.com/problemset/problem/20/C>

In an undirected graph with up to 10^5 vertices and edges and positive edge weights, print one shortest path from vertex 1 to vertex n , or -1 if none exists. The path, not just its distance, is required.

3.6 12. 161D – Distance in Tree (1800)

Official: <https://codeforces.com/problemset/problem/161/D>

For a tree with up to $5 \cdot 10^4$ vertices and a distance $k \leq 500$, count unordered pairs of distinct vertices whose simple-path distance is exactly k . A direct BFS from every vertex is too slow.

4 Gate 3: compressed state and data structures

4.1 13. 580D – Kefa and Dishes (1800)

Official: <https://codeforces.com/problemset/problem/580/D>

Choose exactly m of at most 18 dishes and order the chosen dishes. Each dish has a base satisfaction, and selected ordered pairs add a bonus when the first dish immediately precedes the second. Maximize total satisfaction.

4.2 14. 1245D – Shichikuji and Power Grid (1900)

Official: <https://codeforces.com/problemset/problem/1245/D>

For up to 2000 cities, either build a station in a city or connect cities by wires. Station and Manhattan-distance wire costs are supplied. Make every city receive power at minimum cost, then print the chosen stations and wires.

4.3 15. 271D – Good Substrings (1800)

Official: <https://codeforces.com/problemset/problem/271/D>

Given a lowercase string of length at most 1500, a 26-character good/bad mask, and k , count distinct substrings containing at most k bad characters. Equal substrings from different positions count once.

4.4 16. 61E – Enemy Is Weak (1900)

Official: <https://codeforces.com/problemset/problem/61/E>

For an array of up to 10^6 distinct values, count triples $i < j < k$ with $a[i] > a[j] > a[k]$. The answer can require 64 bits.

4.5 17. 86D – Powerful Array (2200)

Official: <https://codeforces.com/problemset/problem/86/D>

For up to $2 * 10^5$ range queries, compute $\text{sum}(\text{value} * \text{frequency}(\text{value})^2)$ over each subarray. Values are bounded, but recomputing every query from scratch is too slow. Print answers in query order.

5 Gate 4: reductions and optimization

5.1 18. 498C – Array and Operations (2100)

Official: <https://codeforces.com/problemset/problem/498/C>

Given numbers on odd and even indexed vertices and allowed odd-even links, perform as many operations as possible. An operation chooses a linked pair and any common divisor greater than one, then divides both current numbers by that divisor. Find the maximum number of operations.

5.2 19. 319C – Kalila and Dimna in the Logging Industry (2100)

Official: <https://codeforces.com/problemset/problem/319/C>

The official story supplies strictly increasing tree heights a and strictly decreasing recharge costs b , with $n \leq 10^5$. Compute the minimum total cutting cost under the stated transition rules. The answer can exceed 32-bit integer range.

6 Gate 5: hard data structures and tree techniques

6.1 20. 786B – Legacy

Official: <https://codeforces.com/problemset/problem/786/B>

Maintain a directed weighted graph under edge additions from one vertex to one vertex, from one vertex to every vertex in an interval, and from every vertex in an interval to one vertex. Then print shortest distances from a source.

6.2 21. 191C – Fools and Roads

Official: <https://codeforces.com/problemset/problem/191/C>

Given a tree and many vertex pairs, count for every edge how many requested simple paths use it. Print counts in input-edge order.

6.3 22. 208E – Blood Cousins

Official: <https://codeforces.com/problemset/problem/208/E>

In a rooted forest, query a vertex v and integer p : count other vertices at the depth of v whose p -th ancestor equals the p -th ancestor of v .

6.4 23. 375D – Tree and Queries

Official: <https://codeforces.com/problemset/problem/375/D>

Each tree vertex has a color. For every (v, k) , count colors occurring at least k times inside the subtree of v .

6.5 24. 600E – Lomsat gelral

Official: <https://codeforces.com/problemset/problem/600/E>

For every tree vertex, sum all color values whose frequency is maximum in its subtree.

6.6 25. 877E – Danil and a Part-time Job

Official: <https://codeforces.com/problemset/problem/877/E>

Root a tree. Toggle all binary values in a queried subtree or report a queried subtree sum.

6.7 26. 242E – XOR on Segment

Official: <https://codeforces.com/problemset/problem/242/E>

Maintain an array under range xor updates and range-sum queries.

6.8 27. 103D – Time to Raid Cowavans

Official: <https://codeforces.com/problemset/problem/103/D>

For each (a, b) , sum $a[a] + a[a+b] + a[a+2b] + \dots$ while indices remain in bounds.

7 Gate 6: expert modeling and offline structures

7.1 28. 1000F – One Occurrence

Official: <https://codeforces.com/problemset/problem/1000/F>

For every range $[l, r]$, print any value occurring exactly once in that range, or zero when none exists.

7.2 29. 240F – TorCoder

Official: <https://codeforces.com/problemset/problem/240/F>

Maintain a lowercase string while substring operations sort one selected range in increasing or decreasing order. Output the final string.

7.3 30. 343D – Water Tree

Official: <https://codeforces.com/problemset/problem/343/D>

Root a tree and support filling a subtree, emptying a root-to-vertex path, and querying whether one vertex is filled.

7.4 31. 1209E2 – Rotate Columns

Official: <https://codeforces.com/problemset/problem/1209/E2>

Choose cyclic rotations of selected matrix columns so every row takes one value and the resulting total is maximum.

7.5 32. 1416C – XOR Inverse

Official: <https://codeforces.com/problemset/problem/1416/C>

Choose one xor mask for all array values to minimize the transformed inversion count. Output the count and one valid mask.

7.6 33. 13E – Holes

Official: <https://codeforces.com/problemset/problem/13/E>

Array values define jumps $i \rightarrow i+a[i]$. Support point updates and queries for the number of jumps to leave the array and the final in-array position.

7.7 34. 342E – Xenia and Tree

Official: <https://codeforces.com/problemset/problem/342/E>

Initially one tree vertex is red. Repeatedly color vertices red or query the nearest-red distance from a vertex.

7.8 35. 895C – Square Subsets

Official: <https://codeforces.com/problemset/problem/895/C>

Count nonempty subsets of a bounded-value multiset whose product is a perfect square, modulo the required prime.

7.9 36. 868F – Yet Another Minimization Problem

Official: <https://codeforces.com/problemset/problem/868/F>

Partition an array into exactly k nonempty segments. Segment cost is the number of equal-value pairs in it; minimize total cost.

7.10 37. 1292C – Xenon’s Attack on the Gangs

Official: <https://codeforces.com/problemset/problem/1292/C>

Choose two tree vertices and maximize the product of the endpoint component sizes obtained by cutting toward the other endpoint along their path.

7.11 38. 613D – Kingdom and its Cities

Official: <https://codeforces.com/problemset/problem/613/D>

For each marked set of tree vertices, delete as few other vertices as possible so no component contains two marked vertices, or report impossibility.

7.12 39. 1110F – Nearest Leaf

Official: <https://codeforces.com/problemset/problem/1110/F>

For a weighted rooted tree, answer minimum-distance-to-leaf queries restricted to a DFS-order interval of leaves.

7.13 40. 587C – Duff in the Army

Official: <https://codeforces.com/problemset/problem/587/C>

People occupy tree vertices. For each path, print the smallest few person ids on that path, up to the requested limit.

8 Gate 7: red-level optimization

8.1 41. 786C – Till I Collapse

Official: <https://codeforces.com/problemset/problem/786/C>

For every k , partition the array greedily into the fewest segments containing at most k distinct values and print all answers.

8.2 42. 1093G – Multidimensional Queries

Official: <https://codeforces.com/problemset/problem/1093/G>

Maintain points in dimension at most five under point updates; for each index range, output the maximum Manhattan distance between two points in it.

8.3 43. 321E – Ciel and Gondolas

Official: <https://codeforces.com/problemset/problem/321/E>

Partition a binary compatibility matrix into exactly k contiguous groups and minimize the sum of group-square disagreement costs.

8.4 44. 896C – Willem, Chtholly and Seniorious

Official: <https://codeforces.com/problemset/problem/896/C>

Support range assignment, range addition, k th-smallest, and modular power-sum queries on the statement's generated array.

8.5 45. 833B – The Bakery

Official: <https://codeforces.com/problemset/problem/833/B>

Split an array into exactly k nonempty segments. A segment scores its number of distinct values; maximize total score.

8.6 46. 311B – Cats Transport

Official: <https://codeforces.com/problemset/problem/311/B>

Cats arrive at stations on a line. A limited number of buses carry consecutive groups; minimize total waiting time.

8.7 47. 55D – Beautiful Numbers

Official: <https://codeforces.com/problemset/problem/55/D>

For each interval, count positive integers divisible by every nonzero decimal digit they contain.

8.8 48. 840D – Destiny

Official: <https://codeforces.com/problemset/problem/840/D>

For each range and integer k , find a value occurring more than length / k times, or report that no such value exists.

8.9 49. 911G – Mass Change Queries

Official: <https://codeforces.com/problemset/problem/911/G>

Apply range operations that replace every occurrence of one value by another, then print the resulting array.

8.10 50. 1100F – Ivan and Burgers

Official: <https://codeforces.com/problemset/problem/1100/F>

For each array range, choose any subset of its values to maximize their xor.

9 Offline blind capstone

9.1 D. Ribbon Cuts – Codeforces 189A companion

Official companion: <https://codeforces.com/problemset/problem/189/A>

Given a ribbon of length n and three permitted positive piece lengths a , b , and c , use the whole ribbon and maximize the number of pieces. Print -1 when no exact cutting exists. The local version has $n \leq 200000$ and is also available in `problems/d_ribbon_cuts` with fixed tests for offline work.

10 Offline attempt log

For every task, leave one line before opening the editorial:

```
model: _____ invariant: _____  
first hostile test: _____ exact blocker: _____
```